Slip 1

Java

```
import java.util.Arrays;
class ArraySort
{
   public static void main(String[] args)
{
      int a[] = {5, 2, 8, 1, 3};
      int sum = 0;
      for (int x : a)
         sum += x;
      Arrays.sort(a);
      System.out.println("Sum = " + sum);
      System.out.println("Ascending order: "
                  + Arrays.toString(a));
   }
}
```

WT

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Registration</title>
   <style>
      body {
         font-family: Arial;
         background: #ffffe0;
      }
      form {
         width: 400px;
         margin: auto;
         padding: 20px;
         background: #fffde7;
      }
```

```
    h2 {
      text-align: center;
    }
    input, select, textarea {
      margin: 5px;
    }
  </style>
</head>
<body>

<form>
  <h2>My Future College / School</h2>

  Student Name:
  <input type="text" minlength="3" maxlength="10" required><br>

  Date of Birth:
  <input type="date" required><br>

  Mobile No:
  <input type="tel" pattern="[0-9]{10}" required><br>

  Email:
  <input type="email" required><br>

  Password:
  <input type="password" minlength="8" required><br>

  Gender:
  <input type="radio" name="gender" required>Male
  <input type="radio" name="gender">Female<br>

  Hobbies:
  <input type="checkbox">Drawing
  <input type="checkbox">Singing
  <input type="checkbox">Dancing<br>

  Course:
  <select required>
```

```
    <option value="">Select Course</option>
    <option>BSc Computer Science</option>
    <option>BCA</option>
    <option>BCom</option>
  </select><br>

  Student Photo:
  <input type="file" accept="image/*"><br>

  Register Time:
  <input type="time" required><br>

  Address:
  <textarea required></textarea><br>

  <input type="checkbox" required>
  I declare the information is correct.<br><br>

  <input type="submit" value="Register">
  <input type="reset" value="Reset">
</form>
</body>
</html>
```

Slip 2

Java

```
import java.util.Scanner;

class Armstrong {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int a = sc.nextInt(), b = sc.nextInt();

    for (int n = a; n <= b; n++) {
      int t = n, sum = 0;

      while (t > 0) {
        int d = t % 10;
        sum += d * d * d;
```

```
            t /= 10;
        }

        if (sum == n)
            System.out.print(n + " ");
    }
    sc.close();
  }
}
```

WT

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration Form</title>
<style>
body {
  background: #ffe4e1;
  font-family: Arial;
}
form {
  width: 450px;
  margin: auto;
  padding: 20px;
}
h2 {
  text-align: center;
}
input, select, textarea {
  margin: 5px;
}
</style>
</head>
<body>

<form>
<h2>Student Registration Form</h2>
```

```
Roll No:
<input type="text" required><br>

Student Name:
<input type="text" placeholder="First Name" required>
<input type="text" placeholder="Last Name" required><br>

Father's Name:
<input type="text" required><br>

Date of Birth:
<input type="date" required><br>

Mobile No:
<input type="tel" pattern="[0-9]{10}" required><br>

Email:
<input type="email" required><br>

Password:
<input type="password" minlength="8" required><br>

Gender:
<input type="radio" name="gender" required>Male
<input type="radio" name="gender">Female<br>

Department:
<input type="checkbox">CSE
<input type="checkbox">IT
<input type="checkbox">ECE
<input type="checkbox">Civil
<input type="checkbox">Mech<br>

Course:
<select required>
<option value="">Select Course</option>
<option>BCA</option>
<option>BSc CS</option>
<option>BTech</option>
```

```
</select><br>

Student Photo:
<input type="file" accept="image/*"><br>

City:
<input type="text" required><br>

Address:
<textarea required></textarea><br><br>

<input type="submit" value="Register">
<input type="reset" value="Reset">

</form>
</body>
</html>
```

Slip 4

Java

```
import java.util.Scanner;

class Matrix {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int ch, i, j, k;
      int[][] a = new int[10][10];
      int[][] b = new int[10][10];
      int[][] c = new int[10][10];
      int r, col, r2, c2;

      do {
         System.out.println("\n1.Addition");
         System.out.println("2.Multiplication");
         System.out.println("3.Transpose");
         System.out.println("4.Exit");
         System.out.print("Enter choice: ");
```

```
ch = sc.nextInt();

switch (ch) {
   case 1:
      System.out.print("Enter rows and columns: ");
      r = sc.nextInt();
      col = sc.nextInt();

      System.out.println("Enter first matrix:");
      for (i = 0; i < r; i++)
         for (j = 0; j < col; j++)
            a[i][j] = sc.nextInt();

      System.out.println("Enter second matrix:");
      for (i = 0; i < r; i++)
         for (j = 0; j < col; j++)
            b[i][j] = sc.nextInt();

      for (i = 0; i < r; i++) {
         for (j = 0; j < col; j++) {
            c[i][j] = a[i][j] + b[i][j];
            System.out.print(c[i][j] + " ");
         }
         System.out.println();
      }
      break;

   case 2:
      System.out.print("Enter rows and columns of first matrix: ");
      r = sc.nextInt();
      col = sc.nextInt();

      System.out.print("Enter rows and columns of second matrix: ");
      r2 = sc.nextInt();
      c2 = sc.nextInt();

      if (col != r2) {
         System.out.println("Multiplication not possible");
         break;
```

```
    }

    System.out.println("Enter first matrix:");
    for (i = 0; i < r; i++)
        for (j = 0; j < col; j++)
            a[i][j] = sc.nextInt();

    System.out.println("Enter second matrix:");
    for (i = 0; i < r2; i++)
        for (j = 0; j < c2; j++)
            b[i][j] = sc.nextInt();

    for (i = 0; i < r; i++) {
        for (j = 0; j < c2; j++) {
            c[i][j] = 0;
            for (k = 0; k < col; k++)
                c[i][j] += a[i][k] * b[k][j];
            System.out.print(c[i][j] + " ");
        }
        System.out.println();
    }
    break;

case 3:
    System.out.print("Enter rows and columns: ");
    r = sc.nextInt();
    col = sc.nextInt();

    System.out.println("Enter matrix:");
    for (i = 0; i < r; i++)
        for (j = 0; j < col; j++)
            a[i][j] = sc.nextInt();

    System.out.println("Transpose:");
    for (j = 0; j < col; j++) {
        for (i = 0; i < r; i++)
            System.out.print(a[i][j] + " ");
        System.out.println();
    }
```

```
                break;

            case 4:
                System.out.println("Exit");
                break;

            default:
                System.out.println("Invalid choice");
        }
    } while (ch != 4);

    sc.close();
  }
}
```

WT

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Heading</title>
</head>
<body>
  <h1 id="msg">Hello! Welcome</h1>

  <button onclick="changeText()">Click Me</button>

  <script>
    function changeText() {
      document.getElementById("msg").innerHTML =
        "Text Changed Successfully";
    }
  </script>
</body>
</html>
```

Slip 6

Java

```
class Account {
   String custname;
   int accno;

   Account() {
      custname = "ABC";
      accno = 101;
   }

   Account(String n, int a) {
      custname = n;
      accno = a;
   }
}

class SavingAccount extends Account {
   double savingbal = 5000, minbal = 1000;
}

class AccountDetail extends SavingAccount {
   double depositamt = 2000, withdrawalamt = 500;

   void display() {
      System.out.println("Name: " + custname);
      System.out.println("Account No: " + accno);
      System.out.println("Saving Balance: " + savingbal);
      System.out.println("Minimum Balance: " + minbal);
      System.out.println("Deposit: " + depositamt);
      System.out.println("Withdrawal: " + withdrawalamt);
   }

   public static void main(String[] args) {
      AccountDetail a = new AccountDetail();
```

```
        a.display();
    }
}
```

WT

```
<!DOCTYPE html>
<html>
<head>
<style>
button:hover {
    background-color: green;
    color: white;
}
a:hover {
    color: red;
    font-size: 25px;
}
</style>
</head>
<body>
    <button>Click Me</button>
    <br><br>
    <a href="#">Visit Website</a>
</body>
</html>
```

Slip 7

Java

```
class AgeException extends Exception {
    AgeException(String msg) {
        super(msg);
    }
}

class Driver {
    String license_no, name, address;
    int age;
```

```
   Driver(String l, String n, String a, int age)
         throws AgeException {
      if (age < 18)
         throw new AgeException("Age is below 18 years");

      license_no = l;
      name = n;
      address = a;
      this.age = age;
   }

   public static void main(String[] args) {
      try {
         Driver d = new Driver("1234", "Rahul",
                        "Pune", 16);
         System.out.println(d.name);
      } catch (AgeException e) {
         System.out.println(e.getMessage());
      }
   }
}
```

WT

```
<!DOCTYPE html>
<html>
<head>
<style>
li:first-child {
   color: red;
   font-weight: bold;
}
li:last-child {
   color: blue;
   font-style: italic;
}
</style>
</head>
<body>
<h2>Programming Languages</h2>
```

```
<ul>
   <li>Java</li>
   <li>Python</li>
   <li>C</li>
   <li>JavaScript</li>
</ul>
</body>
</html>
```

Slip 8

Java

```
abstract class Shape {
   int a = 10, b = 5;
   abstract void printArea();
}

class Rectangle extends Shape {
   void printArea() {
      System.out.println("Rectangle = " + (a * b));
   }
}

class Triangle extends Shape {
   void printArea() {
      System.out.println("Triangle = " + (0.5 * a * b));
   }
}

class Circle extends Shape {
   void printArea() {
      System.out.println("Circle = " +
         (3.14 * a * a));
   }
}

class Main {
   public static void main(String[] args) {
      new Rectangle().printArea();
```

```
        new Triangle().printArea();
        new Circle().printArea();
    }
}
```

WT

```
<!DOCTYPE html>
<html>
<body>

<h2>Change Image</h2>

<img id="myImage" src="image1.jpg" width="200">

<br><br>

<button onclick="changeImage()">Change Image</button>

<script>
function changeImage() {
    document.getElementById("myImage").src = "image2.jpg";
}
</script>

</body>
</html>
```

Slip 9

Java

```
import java.util.Scanner;

class Employee {
    int id;
    String name;
    double salary;

    void accept(Scanner sc) {
        System.out.print("Enter id, name and salary: ");
```

```
        id = sc.nextInt();
        name = sc.next();
        salary = sc.nextDouble();
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of employees: ");
        int n = sc.nextInt();

        Employee e[] = new Employee[n];

        for (int i = 0; i < n; i++) {
            e[i] = new Employee();
            e[i].accept(sc);
        }

        int max = 0;
        for (int i = 1; i < n; i++) {
            if (e[i].salary > e[max].salary)
                max = i;
        }

        System.out.println("Highest Salary Employee: " + e[max].name);
    }
}
```

WT

```
<!DOCTYPE html>
<html>
<body>
<h3>Arrow Functions</h3>
<script>
let a = 10, b = 5;

let add = (a,b) => a+b;
let sub = (a,b) => a-b;
```

```
let mul = (a,b) => a*b;
let div = (a,b) => a/b;

document.write("Addition: "+add(a,b)+"<br>");
document.write("Subtraction: "+sub(a,b)+"<br>");
document.write("Multiplication: "+mul(a,b)+"<br>");
document.write("Division: "+div(a,b));
</script>
</body>
</html>
```

Slip 10

Java

```
import java.io.*;

class ReadFile {
   public static void main(String[] args) throws Exception {
      FileReader f = new FileReader("abc.txt");
      int ch;

      while ((ch = f.read()) != -1)
         System.out.print(Character.toUpperCase((char)ch));

      f.close();
   }
}
```

WT

```
<!DOCTYPE html>
<html>
<body>
<script>
let name = "Rahul";
let roll = 101;
let marks = 85;

let report = `
Student Report
```

```
Name: ${name}
Roll No: ${roll}
Marks: ${marks}`;

document.write(report.replace(/\n/g, "<br>"));
</script>
</body>
</html>
```

Slip 11

Java

```
import java.util.*;

class Vehicle {
   String company;
   double price;

   void accept(Scanner sc) {
      company = sc.next();
      price = sc.nextDouble();
   }
}

class LightMotorVehicle extends Vehicle {
   int mileage;

   void accept(Scanner sc) {
      super.accept(sc);
      mileage = sc.nextInt();
   }

   void display() {
      System.out.println(company+" "+price+" "+mileage);
   }
}

class HeavyMotorVehicle extends Vehicle {
   int capacity;
```

```
    void accept(Scanner sc) {
        super.accept(sc);
        capacity = sc.nextInt();
    }

    void display() {
        System.out.println(company+" "+price+" "+capacity);
    }
}

class Demo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i=0; i<n; i++) {
            int type = sc.nextInt();
            if(type == 1) {
                LightMotorVehicle l = new LightMotorVehicle();
                l.accept(sc);
                l.display();
            } else {
                HeavyMotorVehicle h = new HeavyMotorVehicle();
                h.accept(sc);
                h.display();
            }
        }
    }
}
```

WT

```
<!DOCTYPE html>
<html>
<body>
<script>
let total = (a, b, c) => a + b + c;

let m1 = 80, m2 = 90, m3 = 70;
```

```
let t = total(m1, m2, m3);
let avg = t / 3;

document.write(`
Total Marks: ${t}<br>
Average Marks: ${avg}
`);
</script>
</body>
</html>
```

Slip 12

Java

1.Indooe.java

```
package game;

public class Indoor {
   String name;

   public Indoor() {
      name = "Chess";
   }

   public Indoor(String n) {
      name = n;
   }

   public void display() {
      System.out.println("Indoor Game: " + name);
   }
}
```

2.Outdoor.java

```
package game;

public class Outdoor {
   String name;

   public Outdoor() {
```

```
        name = "Cricket";
    }

    public Outdoor(String n) {
        name = n;
    }

    public void display() {
        System.out.println("Outdoor Game: " + name);
    }
}
```

3.Test.java

```
import game.*;

class Test {
    public static void main(String[] args) {
        Indoor i = new Indoor();
        Outdoor o = new Outdoor();

        i.display();
        o.display();

        Indoor i1 = new Indoor("Carrom");
        Outdoor o1 = new Outdoor("Football");

        i1.display();
        o1.display();
    }
}
```

WT

```
<!DOCTYPE html>
<html>
<body>
<script>
let price = 500, qty = 3;
let total = (p, q) => p * q;
```

```
let bill = `
Product Price: Rs.${price}
Quantity: ${qty}
Total Price: Rs.${total(price, qty)}`;
document.write(bill.replace(/\n/g, "<br>"));
</script>
</body>
</html>
```

Slip 13

Java

```
class Clock {
   int h, m, s;

   Clock(int h, int m, int s) {
      if (h >= 1 && h <= 12 && m >= 0 &&
         m <= 59 && s >= 0 && s <= 59) {
         this.h = h;
         this.m = m;
         this.s = s;
      } else {
         System.out.println("Invalid time");
      }
   }

   void display(String mode) {
      System.out.println(h + ":" + m + ":" + s
                  + " " + mode);
   }

   public static void main(String[] args) {
      Clock c = new Clock(10, 30, 45);
      c.display("AM");
   }
}
```

WT

```
<!DOCTYPE html>
<html>
```

```
<body>
<script>
let a = 10, b = 20;

[a, b] = [b, a];

document.write("After swapping:<br>");
document.write("a = " + a + "<br>");
document.write("b = " + b);
</script>
</body>
</html>
```